

If $f : [0, \infty) \rightarrow \mathbb{R}$ is $\text{l\`a}d\text{l\`a}g$, we will denote by f_- the left limit of f (with the convention that $f_{0-} = f_0$), and by f_+ the right limit of f . The left jump of f is denoted by $\Delta f = f - f_-$. For later use, we will state Froda's theorem (in a slightly more general form, applying to finite variation functions of time).

Lemma 1. *Suppose f has locally finite variation. Then f is $\text{l\`a}d\text{l\`a}g$ and*

$$\{t \in [0, \infty) : f \text{ is discontinuous at } t\} = \{f \neq f_-\} \cup \{f \neq f_+\}.$$

Furthermore, the above sets are all at most countable.

3. MAIN RESULTS

The main theorem of this article, Theorem 1, can be stated as follows.

Theorem 1. *Let $(X^n)_{n=1}^\infty$ be a sequence of semimartingales such that, for each $t \geq 0$, the set*

$$\text{co} \left\{ \left| \int_0^t \xi dX^n \right| : n \in \mathbb{N}, \xi \text{ is predictable and } |\xi| \leq 1 \right\}$$

is bounded in probability, and $(X_0^n)_{n=1}^\infty$ is bounded. Then there exists $\tilde{X}^n \in \text{co}\{X^m : m \geq n\}$ and a semimartingale $\tilde{X}$ such that $\tilde{X} = \lim_{s \downarrow \cdot, s \in \mathbb{Q}_+} \lim_{n \rightarrow \infty} \tilde{X}_s^n$ and

$$\lim_{s \downarrow \cdot, s \in \mathbb{Q}_+} \mathbb{P} - \lim_{n \rightarrow \infty} \int_0^s Y d\tilde{X}^n = \int_0^\cdot Y d\tilde{X}$$

for each continuous semimartingale Y .

The continuity condition on Y situates Theorem 1 as a stochastic counterpart to the following well-known version of Prokhorov's theorem: if $(g_n)_{n=1}^\infty$ is a sequence of finite variation functions on $[0, 1]$ with $\sup_n \sup_{t \in [0, 1]} \text{var}(g_n)_t \leq 1$, there exists a subsequence $(n_k)_{k=1}^\infty$ and a finite variation g such that $\lim_{k \rightarrow \infty} \int_0^1 f dg_{n_k} = \int_0^1 f dg$ for all continuous f . In Section 5, we show the continuity condition in Theorem 1 cannot be weakened—mirroring the deterministic case.

The imposition of a boundedness condition on Theorem 1 is natural, and not a significant restriction. Indeed, the boundedness conditions assumed in the literature are often stronger (see, for example, [KP91; DS99]). Furthermore, since the boundedness condition from Theorem 1 is based on boundedness in probability, Theorem 1 is not chained to any particular choice of measure, an important property when working with multiple equivalent probability measures.

Let us note the following corollary of Theorem 1 for H^1 -bounded sequences of martingales.

Corollary 1. *Let $(M^n)_{n=1}^\infty$ be a sequence of martingales such that*

$$\sup_n \int_\Omega (M^n)_t^* d\mathbb{P} < \infty$$